Permanent and Transitory Unobserved Heterogeneity in Credit Ratings

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Abstract

Many central banks have developed systems to produce Point-In-Time (PIT) credit ratings for monetary policy operations, but these ratings are often too sensitive to business cycle conditions and short-term fluctuations for regulatory purposes. This thesis proposes a method to derive more stable Through-The-Cycle (TTC) ratings from PIT ratings by decomposing the latter into persistent and transitory components. The model exploits panel data techniques to account for unobserved heterogeneity in firms' creditworthiness—an aspect often overlooked in existing approaches. Applied to a novel dataset of internal ratings from the Bank of Spain, the resulting TTC ratings are substantially more stable while remaining responsive to structural changes in firms' risk profiles. The estimated year effects trace key downturns—such as the Great Financial Crisis, the Euro Area debt crisis, and the COVID-19 shock—highlighting the strongly procyclical and volatile nature of PIT ratings.

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1 Introduction

Credit ratings are widely used to quantify the creditworthiness of borrowers by summarizing default risk into an ordered categorical scale. These ratings reflect the estimated probability of default and are typically based on historical information about the borrower's financial performance, borrowing behavior, and repayment history, often benchmarked against comparable entities. They are a key tool in credit markets, used to assess the credit risk of sovereign bonds, corporate debt, consumer loans, and commercial lending to firms.

A key distinction in the credit ratings landscape lies between Point-In-Time (PIT) and Through-The-Cycle (TTC) ratings. PIT ratings are designed to respond quickly to changes in firm-specific or macroeconomic conditions and are commonly used for purposes such as determining collateral eligibility. In contrast, TTC ratings aim to abstract from short-term volatility and cyclical fluctuations, providing a more stable assessment of long-term credit risk. This stability makes TTC ratings particularly well suited for regulatory applications like capital requirement setting and internal model validation, where procyclicality is a concern.

Although some Central Banks have already developed robust infrastructures to generate PIT ratings for non-financial firms, little attention has been paid to the derivation of TTC ratings from this existing framework. If PIT ratings could be smoothed to isolate their persistent component, Central Banks could easily produce TTC-style measures. This would significantly expand their regulatory toolkit: for instance, by providing an independent benchmark to assess private banks' internal models or by directly informing macroprudential capital buffers.

This paper proposes a method to extract TTC ratings from PIT ratings by smoothing out transitory components and isolating persistent, firm-specific creditworthiness. With this aim, the smoother is based on an ordered probit version of the canonical permanent-transitory model of income shocks. Given its non-linear nature, the model is estimated using a Stochastic Expectation-Maximization (SEM) algorithm and Markov Chain Monte Carlo (MCMC) methods.

The key innovation of this approach is the use of panel data methods to fully exploit the entire history of a firm's creditworthiness when estimating its TTC rating, accounting for unobserved heterogeneity in a firm's credit risk profile. In contrast, most existing methods for deriving TTC ratings from PIT ratings rely solely on the last observed rating. For example, the standard practice in private banks' internal ratings is to scale the estimated probabilities of default (PDs) by a constant factor so that the portfolio's average PD matches the historical

average default rate[1]. Such approaches overlook the valuable information embedded in the full time series of PIT ratings.

To implement this approach, I first study a novel dataset of credit ratings assigned by Bank of Spain to Spanish non-financial firms—including micro, small and medium-sized enterprises—covering the period from 2004 to 2023. Their implied transition matrices reveal that these ratings are clearly aligned with the PIT philosophy, as expected given their role in determining collateral eligibility within monetary policy operations.

The estimated TTC ratings exhibit considerably stronger persistence than the original PIT ratings, resulting in a more stable distribution of credit ratings over the business cycle. By construction, they smooth out transitory shocks to a firm's creditworthiness, including short-term fluctuations caused by temporary disruptions such as supply chain issues, machine breakdowns, or delays in customer payments. Additionally, TTC ratings attenuate the influence of macroeconomic conditions on firms' risk profiles, reflecting long-term fundamentals rather than cyclical volatility.

Importantly, the cyclical component embedded in PIT ratings is not imposed ex-ante but emerges endogenously from the data through the estimation of year fixed effects. These fixed effects closely track macroeconomic dynamics, indicating a deterioration in credit quality during the Great Financial Crisis, the European sovereign debt crisis and the COVID-19 pandemic in 2020, and an improvement during subsequent recovery periods. The model thus captures cyclical behavior without explicitly labeling periods as recessions or expansions. This further underscores the procyclical nature of PIT ratings, in contrast to the TTC ratings, which remain comparatively stable throughout the cycle.

This thesis contributes to two main strands of the credit ratings literature. The first examines the cyclical behavior of credit ratings and their sensitivity to macroeconomic conditions. The second focuses on the distinction between PIT and TTC ratings, particularly in regulatory contexts.

In the first strand, previous research has studied ratings cyclicality by comparing transition matrices estimated separately for expansionary and recessionary periods. Seminal contribu-

[1]The typical functioning of private banks' Internal Ratings-Based approaches is as follows. First, banks estimate credit risk models to support loan decisions, focusing on predictive performance rather than explicitly targeting PIT or TTC characteristics. The resulting models generate credit scores, which are then mapped to PDs. The cyclical behavior of these models is assessed ex post and often varies by loan type—for instance, mortgage models tend to behave more like TTC. Second, for capital requirement calculations, banks calibrate the estimated PDs so that the average PD of the portfolio aligns with its historical observed default rates. A common approach is to apply a constant scaling factor across all PDs, though more advanced methods allow the scaling factor to vary with the underlying score. Intuitively, if the model is estimated during an expansion, raw PDs may be too low, and scaling adjusts them upward to reflect long-run credit risk.

tions include Nickell et al. (2000) and Bangia et al. (2002). The former find that migration probabilities—and thus implied PDs—are highly sensitive to the business cycle, especially for lower-rated firms. The latter show that first-order transition matrices approximate ratings dynamics reasonably well only when regime-switching models account for the business cycle. More recently, Gredil et al. (2022) argue that ratings are more informative during recessions than in booms and that an essential function of Credit Rating Agencies is to distinguish between transitory and permanent credit shocks.

Unlike most prior work, I do not impose any ex-ante definition of expansions or recessions. Instead, the model endogenously captures the cyclical component through estimated year fixed effects. This avoids relying on subjective or potentially misaligned macroeconomic classifications. For example, the Business Cycle Dating Committee of the Spanish Economic Association classifies 2014 and 2015 as expansion years, but my estimated year fixed effects suggest those years still exhibited deteriorating credit quality, likely as a lingering consequence of the sovereign debt crisis. This underscores the value of allowing the data to reveal cyclical dynamics, rather than imposing them externally.

In the second strand, the literature explores how to reconcile PIT and TTC ratings. Repullo et al. (2010) examine strategies to reduce the procyclicality of capital requirements under Basel II, concluding that either smoothing capital buffers or using TTC PDs can be effective, with the former offering greater transparency and ease of implementation. Other studies focus on deriving one type of rating from the other. For instance, Aguais et al. (2004) propose a method to extract PIT ratings from the TTC ratings published by Credit Rating Agencies, while Aguais et al. (2006) develop a unified framework capable of generating both PIT and TTC ratings by adjusting for the business cycle. More recently, Fidalgo et al. (2024) introduce a method to estimate TTC scores from PIT scores by subtracting the cyclical component —measured through a regression of PIT scores on macroeconomic variables. These approaches, however, primarily focus on macroeconomic adjustments and do not account for firm-specific transitory shocks that may temporarily distort credit risk assessments.

This thesis contributes to the literature by proposing a novel method that fully exploits the panel structure of PIT ratings to recover TTC ratings. Unlike existing approaches that rely solely on the most recent observed rating, the model uses the entire time series of each firm's credit history to disentangle persistent from transitory components of creditworthiness. This is particularly important when recent rating movements reflect temporary disruptions —such as liquidity shortages or operational delays— rather than long-term deterioration. Fur-

thermore, the model adjusts flexibly for macroeconomic conditions without imposing explicit relationships with specific indicators, allowing the cyclical component to emerge directly from the data.

The thesis is organized as follows. Section 2 describes the dataset from Bank of Spain, showing how ratings dynamics suggest that these are PIT ratings. Section 3 introduces the model and outlines the process for constructing TTC ratings from PIT ratings. Section 4 then applies the model to the PIT ratings of Spanish non-financial firms, estimating TTC ratings and analyzing their cyclical properties.

2 Data

This thesis draws on a novel dataset of internal credit ratings for Spanish non-financial firms, compiled by the Bank of Spain. The data cover the period from 2000 to 2023 and include firms of all sizes—from large corporations to microenterprises. While credit ratings are available throughout the entire period, information on defaults and insolvency proceedings is only available from 2004 onward. For this reason, the analysis focuses on the 2004–2023 period, during which the dataset comprises 17,299,987 firm-year observations for 2,199,752 unique firms.

Each firm-year observation includes: (i) the internal credit rating assigned by the Bank of Spain, (ii) the firm's economic sector and size classification[2], (iii) indicators for current default or insolvency proceedings, (iv) an indicator for whether the firm initiates insolvency proceedings in the subsequent year, and (v) the percentage of exposure in default during the following year, as reported in the Central Credit Register (CCR).

The ratings are generated using a logistic model based on firms' balance sheet data. The estimated scores are mapped into 19 ordered rating categories using historical default rates (from best to worst: $1, 2+, 2, 2-, \ldots, 7+, 7, 7-$). However, only a subset of these ratings appears in the data, specifically from 4+ to 7−. For large publicly listed firms, a rating analyst reviews the model-based rating and may either confirm or override it.

Although the statistical model does not include macroeconomic variables explicitly, it indirectly captures the state of the economy by relying on recent firm-level balance sheet data. The two main data sources are the Central Balance Sheet Data Office (CBI), an internal database from Bank of Spain containing financial statements filed with the Mercantile

[2]There are 34 economic sectors, defined according to the National Classification of Economic Activities (CNAE) 2009, and four size categories (micro, small, medium and large), determined following the European Commission's Recommendation 2003/361/EC.

	4+	4	4-	5+	5	5-	6+	6	6-	7+	7	7-	D
4+	**0.45**	0.39	0.1	0.02	0.01	0.01	0.01	0	0	0	0	0	0.00
4	0	**0.73**	0	0.16	0.01	0.03	0.03	0.02	0	0.01	0	0	0.00
4-	0.01	0.16	**0.54**	0.17	0.04	0.03	0.01	0.01	0	0	0	0	0.01
5+	0	0.06	0	**0.62**	0.15	0.02	0.11	0.02	0.01	0.01	0	0	0.00
5	0	0	0	0.32	**0.31**	0	0.26	0.04	0.03	0.01	0.01	0	0.00
5-	0	0.14	0	0.28	0.02	**0.2**	0.18	0.08	0.01	0.06	0.01	0.01	0.01
6+	0	0.01	0	0.13	0.17	0.01	**0.44**	0.09	0.07	0.04	0.02	0	0.01
6	0	0.01	0	0.05	0.08	0.02	0.36	**0.18**	0.13	0.09	0.05	0.01	0.01
6-	0	0	0	0.03	0.06	0	0.26	0.15	**0.28**	0.12	0.09	0.01	0.02
7+	0	0.01	0	0.02	0.03	0.01	0.16	0.12	0.17	**0.29**	0.13	0.03	0.03
7	0	0	0	0.01	0.02	0	0.1	0.07	0.13	0.19	**0.41**	0.03	0.04
7-	0	0	0	0.01	0	0.01	0.06	0.07	0.05	0.17	0.31	**0.17**	0.13

	4+	4	4-	5+	5	5-	6+	6	6-	7+	7	7-	D
4+	**0.35**	0.42	0.12	0.05	0.01	0.02	0.01	0.01	0	0.01	0	0	0.01
4	0	**0.68**	0	0.17	0.01	0.04	0.05	0.02	0	0.01	0	0	0.00
4-	0.02	0.19	**0.43**	0.18	0.05	0.05	0.02	0.01	0	0.01	0	0	0.03
5+	0	0.07	0	**0.57**	0.15	0.02	0.13	0.02	0.01	0.01	0.01	0	0.01
5	0	0.01	0	0.33	**0.27**	0	0.26	0.04	0.04	0.02	0.01	0	0.01
5-	0	0.16	0	0.26	0.03	**0.16**	0.17	0.08	0.01	0.06	0.01	0.02	0.03
6+	0	0.01	0	0.15	0.17	0.01	**0.39**	0.09	0.08	0.05	0.03	0.01	0.02
6	0	0.01	0	0.08	0.09	0.02	0.33	**0.16**	0.12	0.09	0.06	0.01	0.03
6-	0	0	0	0.05	0.08	0	0.26	0.13	**0.24**	0.11	0.09	0.01	0.04
7+	0	0.01	0	0.04	0.04	0.01	0.18	0.11	0.15	**0.25**	0.12	0.02	0.06
7	0	0	0	0.02	0.03	0	0.13	0.07	0.13	0.17	**0.35**	0.03	0.08
7-	0	0.01	0	0.02	0.01	0.01	0.09	0.07	0.07	0.15	0.26	**0.1**	0.22

Table 1: First-order (top) and second-order (bottom) transition matrices. Rows denote the current category; columns indicate the next category. Element (i, j) is estimated as the number of transitions from category in row i to category in column j divided by the total number of transitions from the category in row i to any other category.

Registry, and INFORMA, a private data provider offering financial and payment information on Spanish firms.

Ratings are assigned to all Spanish non-financial firms with valid balance sheet data, irrespective of firm size or public listing status. Validity is determined according to internal Eurosystem rules. For example, financial statements must not be older than two years at the time of rating assignment.

Table 1 reports the first- and second-order transition matrices. These are constructed assuming a discrete multinomial model: I take the number of transitions from a category i to a category j and divide it by the total number of transitions from the category i to any other category. The first-order transition matrix shows the probabilities of moving from one rating category to another in one year, while the second-order transition matrix captures the

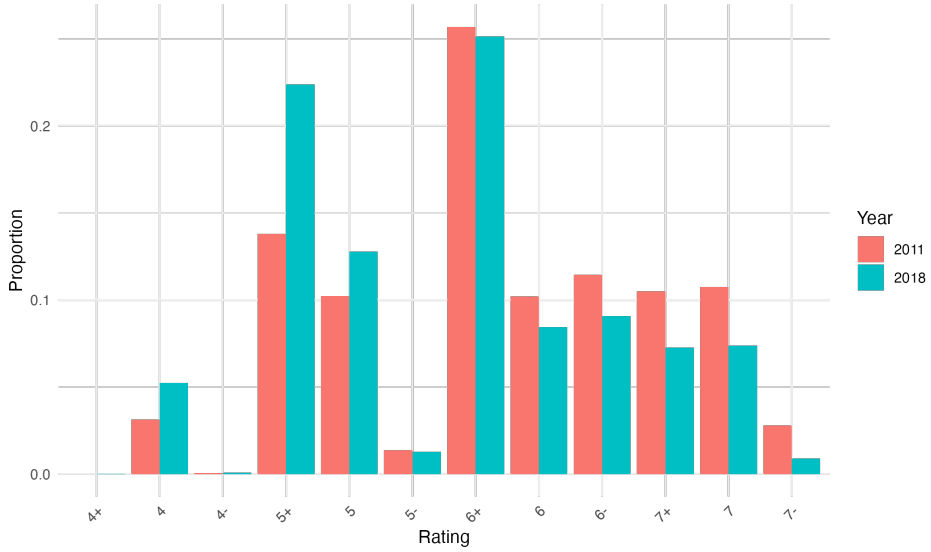


Figure 1: Distribution of credit ratings in 2011 and 2018. The figure shows the proportion of firms in each rating category.

probabilities of moving from one category to another over two years.

As commonly found in the credit ratings literature, diagonal elements generally dominate. However, they are not particularly high. This suggests that the ratings are relatively sensitive to changes in firms' conditions –consistent with a PIT approach. This is expected, given that the primary function of these ratings is to assess asset eligibility for use as collateral in monetary policy operations.

The second-order transition matrix shows limited decay in the diagonal elements, indicating persistence in rating dynamics. This persistence can be exploited to extract the underlying TTC component of creditworthiness, that smoothes transitory shocks as well as business cycle effects.

Figure 1 illustrates the evolution of the distribution of ratings over the business cycle[3]. During 2011, a period of economic distress linked to the European sovereign debt crisis, the distribution shifts toward lower rating categories, indicating a deterioration in aggregate credit quality. In contrast, the distribution in 2018, a period of economic recovery, shifts toward higher categories. These patterns further confirm the PIT nature of the ratings, as they react strongly to macroeconomic conditions.

Due to computational constraints, the analysis in the following sections focuses on a random subsample of 5,000 firms. This subsample is drawn from the pool of firms observed in

[3]Given the low number of large firms, figure 6 in the Appendix A shows the same patterns hold for a sample restricted to large firms, offering a clearer view of the dynamics in higher rating categories, where large firms are more concentrated.

consecutive years and present in both 2004 and 2005, ensuring at least one transition can be observed. To ensure enough transitions are observed for estimation, I merge adjacent rating categories with limited observations: ratings 4+ and 4− are recoded as 4, and 7− is recoded as 7. In addition, I define two new categories:

- Default (D): Indicates that the firm is currently in default.
- Bankruptcy (Q): Assigned to the final observation of a firm when there is a negative signal, i.e., the firm is either in default or undergoing insolvency proceedings, currently or in the following year. This absorbing category captures firms that exit the sample as a result of financial distress.

Table 4 in the Appendix A shows the number of firm-year observations in each rating category for the random subsample, while table 5 shows the same for the full sample.

Firms can exit the sample in two ways: either they are classified as bankrupt (category Q), or they simply stop being observed without a negative signal. The former is modeled as an absorbing state. The latter is assumed to be non-informative, a standard assumption in the credit ratings literature. There are many potential reasons for such non-informative exits —e.g., firm closure due to owner retirement, firm acquisition, or gaps in balance sheet reporting driven by formal issues. Given that such exits can be driven by positive or neutral as well as negative causes, the usual approach is not to extract information from them[4].

3 Methodology

3.1 Model

Let $i \in \{1, \ldots, n\}$ index firms, $t \in \{1, \ldots, T(i)\}$ index time periods, and $j \in \mathcal{G} := \{1, 2, \ldots, G\}$ denote ordered credit rating categories. The model relies on a latent continuous variable y_{it}^* that captures the underlying credit quality of firm i at time t. This latent variable consists of a firm-specific component and a time-specific fixed effect. The firm-specific component is the sum of a random walk and a transitory shock. The observed PIT rating, denoted y_{it}^{PIT}, is a discrete variable that reflects the interval in which y_{it}^* falls, according to a

[4]These exits are treated analogously to firms observed in all years. For instance, a firm that is always observed will have ratings from 2004 to 2023. However, there is no transition to study from 2023 to 2024 because this year has not been observed. Non-bankrupt exits will be treated in the same way.

set of thresholds $\{\alpha_j\}$, where $\alpha_0 = -\infty$ and $\alpha_G = \infty$:

$$y_{it}^* = \eta_{it} + \delta_t + \varepsilon_{it}, \tag{1}$$

$$\eta_{it} = \eta_{it-1} + \nu_{it}, \tag{2}$$

$$y_{it}^{PIT} = j \quad \text{if } \alpha_{j-1} \leq y_{it}^* < \alpha_j. \tag{3}$$

The stochastic terms η_{i0}, ε_{it}, and ν_{it} are assumed to be mutually independent across both i and t, and normally distributed with variances σ_η^2, σ_ε^2, and σ_ν^2, respectively.

To clarify the model's structure and interpretation, consider the following examples. Suppose there are three rating categories, with thresholds $\alpha_1 = 0$ and $\alpha_2 = 1$, and assume for simplicity that the year fixed effects are zero, i.e., $\delta_1 = \delta_2 = 0$. Consider two identical firms, 1 and 2, in period $t = 1$, both with $\eta_{11} = \eta_{21} = 0.5$ and $\varepsilon_{11} = \varepsilon_{21} = 0$. Accordingly, both are assigned to rating category 2 at $t = 1$.

In period $t = 2$, suppose firm 1's creditworthiness remains unchanged, while firm 2 experiences a temporary deterioration, such as $\varepsilon_{22} = 0.5$, due to a short-term event like a supply chain disruption or a delay in customer payments. As a result, firm 1 remains in category 2, while firm 2 is downgraded to category 3. Because this deterioration is purely transitory, a TTC rating should ideally disregard such temporary deviations and assign the same rating to both firms, reflecting their identical long-term creditworthiness. Using the persistent component η_{21} as the basis for classification, firm 2 would remain in category 2.

Next, consider how year fixed effects operate in the model by focusing again on firm 1. The fixed effects δ_t shift the location of the thresholds in each period, while preserving the length of the intervals. Specifically, note that:

$$\alpha_{j-1} \leq \eta_{it} + \varepsilon_{it} + \delta_t < \alpha_j \quad \Leftrightarrow \quad \alpha_{j-1,t}^\ddagger \leq y_{it}^\ddagger < \alpha_{j,t}^\ddagger, \tag{4}$$

where $y_{it}^\ddagger := \eta_{it} + \varepsilon_{it}$ and $\alpha_{j,t}^\ddagger := \alpha_j - \delta_t$. In this representation, the thresholds vary across time, while maintaining constant distance between them. Hence, the model can be interpreted as checking whether the idiosyncratic term $y_{it}^\ddagger$ falls within these time-adjusted intervals. Importantly, even though the year fixed effects shift the thresholds linearly, their impact on the probability of being assigned to a rating category is nonlinear, due to the transformation by the cumulative distribution function of the normal distribution.

Now suppose that $t = 2$ corresponds to a recession, captured by an increase in the fixed effect to $\delta_2 = 1$. The adjusted thresholds become $\alpha_{1,2}^\ddagger = -1$ and $\alpha_{2,2}^\ddagger = 0$. Even if $y_{12}^\ddagger = 0.5$

remains the same, the shift in thresholds causes firm 1 to be downgraded to category 3. A TTC rating, however, would aim to abstract from this macroeconomic shock and classify the firm based on the persistent component η_{12} with time-invariant thresholds (thresholds that remain constant through the business cycle), thereby keeping firm 2 in category 2.

Although η_{it} is unobserved, we can estimate its conditional expectation given the observed PIT ratings. This leads to the construction of a TTC rating defined as:

$$y_{it}^{TTC} = j \quad \Leftrightarrow \quad \overline{\alpha}_{j-1} \leq \mathbb{E}[\eta_{it} \mid y_{i1}^{PIT}, \ldots, y_{iT(i)}^{PIT}] < \overline{\alpha}_j, \tag{5}$$

where $\overline{\alpha}_j := \alpha_j - \frac{1}{T}\sum_{t=1}^{T} \delta_t$ are average-adjusted thresholds that remove the influence of the cyclical component captured by the year fixed effects.

3.2 Identification and estimation

To identify the model, I normalize the mean and variance of the initial latent component η_{i0} to 0 and 1, respectively, and set the first threshold to $\alpha_1 = 0$[5]. These restrictions are necessary because the model is only identified up to location and scale shifts. To see this, consider the following reparameterization of the model:

$$\begin{aligned}
y_{it}^{*} &= \eta_{it}^{\dagger} + \delta_t + \sqrt{\sigma_{\varepsilon}^2}\varepsilon_{it}^{\dagger}, \\
\eta_{it}^{\dagger} &= \eta_{it-1}^{\dagger} + \sqrt{\sigma_{\nu}^2}\nu_{it}^{\dagger}, \\
\eta_{i1}^{\dagger} &= \overline{\eta}_0^{\dagger} + \sqrt{\sigma_{\eta}^2}\eta_{i0}^{\dagger} + \sqrt{\sigma_{\nu}^2}\nu_{i1}^{\dagger},
\end{aligned}$$

where $\eta_{i0}^{\dagger}$, $\varepsilon_{it}^{\dagger}$, and $\nu_{it}^{\dagger}$ are independent standard normal variables. In this formulation, the observed category y_{it}^{PIT} is j if and only if:

$$\alpha_{j-1} \leq \overline{\eta}_0^{\dagger} + \sqrt{\sigma_{\eta}^2}\eta_{i0}^{\dagger} + \delta_t + \sqrt{\sigma_{\nu}^2}\sum_{s=1}^{t}\nu_{is}^{\dagger} + \sqrt{\sigma_{\varepsilon}^2}\varepsilon_{it}^{\dagger} < \alpha_j.$$

This expression shows that adding or subtracting a constant from $\overline{\eta}_0^{\dagger}$ or the fixed effects δ_t, and shifting all thresholds α_j accordingly, leaves the inequality unchanged. Thus, to ensure identification, it is necessary to fix both $\overline{\eta}_0^{\dagger} = 0$ and $\alpha_1 = 0$.[6] Similarly, because the inequality is preserved under multiplication by any positive constant, the model is not identified in scale.

[5] Alternatively, one could normalize by setting the mean of the year fixed effects to zero, thereby avoiding threshold adjustments when computing the TTC rating as shown in equation 5.

[6] In the version of the model without year fixed effects, only one normalization is required, so I normalize $\overline{\eta}_0^{\dagger} = 0$. In the model with δ_t, two normalizations are required since any constant added to $\overline{\eta}_0^{\dagger}$ and subtracted from the year fixed effects leaves the expressions unchanged.

Therefore, I normalize the variance of the initial condition to $\sigma_\eta^2 = 1$.

I estimate the model using the SEM algorithm. A key requirement for applying this algorithm is that the integration range of the latent variable y_{it}^* must be independent of the parameters to be estimated. To ensure this, I follow the reparameterization proposed by Ruud (1991), defining a transformed latent variable $\tilde{y}_{it}$ such that its support is parameter-free:

$$\tilde{y}_{it} := \frac{y_{it}^* - \alpha_{j-1}}{\alpha_j - \alpha_{j-1}}, \quad \text{for } y_{it}^{PIT} = j \text{ and } j \in \{2, \ldots, G-1\}, \tag{6}$$

$$\tilde{y}_{it} := y_{it}^* - \alpha_1, \quad \text{for } y_{it}^{PIT} = 1, \tag{7}$$

$$\tilde{y}_{it} := y_{it}^* - \alpha_{G-1}, \quad \text{for } y_{it}^{PIT} = G. \tag{8}$$

Under this transformation, the latent variable $\tilde{y}_{it}$ satisfies:

$$\tilde{y}_{it} \in [0,1) \iff \alpha_{j-1} \le y_{it}^* < \alpha_j, \quad \text{for } j \in \{2, \ldots, G-1\}, \tag{9}$$

$$\tilde{y}_{it} \in (-\infty, 0) \iff y_{it}^* < \alpha_1, \tag{10}$$

$$\tilde{y}_{it} \in [0,\infty) \iff y_{it}^* \ge \alpha_{G-1}. \tag{11}$$

This transformation allows us to write the complete-data log-likelihood in a form where the integration limits are fixed. Define the length of each interval as $\lambda_j := \alpha_j - \alpha_{j-1}$ for $j \in \{2, \ldots, G-1\}$, and set $\lambda_1 = \lambda_G = 1$ for notational convenience. Let $\overline{T(i)} := \frac{1}{n}\sum_{i=1}^n T(i)$ denote the average number of time periods observed across firms. Also define an auxiliary variable I_{it} that represents the admissible support of $\tilde{y}_{it}$ based on the observed rating y_{it}^{PIT}:

$$I_{it} = \begin{cases} (-\infty, 0) & \text{if } y_{it}^{PIT} = 1, \\ [0, \infty) & \text{if } y_{it}^{PIT} = G, \\ [0, 1) & \text{otherwise.} \end{cases}$$

Then, up to a constant, the complete-data log-likelihood takes the following form:

$$\begin{aligned} -2\,\ell(\tilde{y}, y, \eta) = \text{cte} &+ n\log(\sigma_\eta^2) + \frac{1}{\sigma_\eta^2}\sum_{i=1}^n \eta_{i0}^2 + n\,\overline{T(i)}\log(\sigma_\nu^2) \\ &+ \frac{1}{\sigma_\nu^2}\sum_{i=1}^n \sum_{t=1}^{T(i)} (\eta_{it} - \eta_{it-1})^2 + n\,\overline{T(i)}\log(\sigma_\varepsilon^2) \\ &+ \sum_{j=1}^G \sum_{(i,t):y_{it}^{PIT}=j} \left[\frac{1}{\sigma_\varepsilon^2}\left(\lambda_j \tilde{y}_{it} - \eta_{it} + \alpha_1 + \sum_{g=2}^{j-1}\alpha_g\right)^2 - 2\log(\lambda_j) \right] \\ &+ \sum_{i=1}^n \sum_{t=1}^{T(i)} \log\left(\mathbb{1}\{\tilde{y}_{it} \in I_{it}\}\right). \end{aligned} \tag{12}$$

In the Expectation step, I compute the conditional expectation of the complete-data log-likelihood given the observed ratings y^{PIT}. Because obtaining a closed-form solution for the conditional likelihood is analytically intractable, I approximate it using simulation methods. Specifically, I employ an MCMC approach based on a random walk Metropolis-Hastings algorithm. The variance of the proposal distribution is tuned to maintain an acceptance rate around 0.3.

A key advantage of this step is that it can be parallelized across firms, since each firm's contribution to the log-likelihood is independent from the others. This holds under the assumption that year fixed effects are treated as parameters. If, instead, the year fixed effects were modeled as stochastic variables, one would need to specify their marginal distribution and work with the conditional likelihood of the firm-specific latent terms given the year fixed effects. Parallelization in that case becomes more complex[7], which is why I treat the year fixed effects as parameters. Nonetheless, treating them as stochastic variables could be preferable in certain applications –for instance, if one wishes to impose a structure such as an autoregressive process.

In the Maximization step, I maximize the conditional expectation of the complete-data log-likelihood with respect to the model parameters. To avoid solving a high-dimensional joint optimization problem, I adopt the conditional M-steps strategy proposed by Meng and Rubin (1993), updating subsets of parameters conditionally on the others. Let $\lambda^{(s)}$, $\delta^{(s)}$, and $\sigma^{(s)}$ denote the parameter values at iteration s, and define $N := \sum_{i=1}^{n} T(i)$ as the total number of firm-year observations, and n_τ as the number of observations in period τ. The parameter updates are as follows:

$$\sigma_\nu^{2\,(s+1)} = \frac{1}{N} \sum_{i=1}^{n} \sum_{t=1}^{T(i)} \mathbb{E}\left[(\eta_{it} - \eta_{it-1})^2 \,\middle|\, y \right] \tag{13}$$

$$\sum_{(i,t):y_{it}^{PIT}=k} \left\{ -\frac{1}{\lambda_k^{(s+1)} \sigma_\varepsilon^{2\,(s)}} + \mathbb{E}\left[\tilde{y}_{it} \left(\lambda_k^{(s+1)} \tilde{y}_{it} - \eta_{it} - \delta_t^{(s)} + \alpha_1^{(s+1)} + \sum_{g=2}^{k-1} \lambda_g^{(s+1)} \right) \,\middle|\, y \right] \right\}$$
$$+ \sum_{j>k} \sum_{(i,t):y_{it}^{PIT}=j} \mathbb{E}\left[\left(\lambda_j^{(s)} \tilde{y}_{it} - \eta_{it} - \delta_t^{(s)} + \alpha_1^{(s+1)} + \sum_{g=2}^{k} \lambda_g^{(s+1)} + \sum_{g=k+1}^{j-1} \lambda_g^{(s)} \right) \,\middle|\, y \right] = 0 \tag{14}$$

[7] This would involve a two-level loop: an outer loop over the year fixed effects and an inner loop over firm-specific latent variables conditional on the draws of the year effects. While the inner loop could still be parallelized across firms, it might be more efficient to use a single draw per iteration rather than waiting for convergence.

$$\sigma_{\varepsilon}^{2\,(s+1)} = \frac{1}{N}\sum_{j=1}^{G}\sum_{(i,t):y_{it}^{PIT}=j}\mathbb{E}\left[\left(\lambda_j^{(s+1)}\tilde{y}_{it} - \eta_{it} - \delta_t^{(s)} + \alpha_1^{(s+1)} + \sum_{g=2}^{j-1}\lambda_g^{(s+1)}\right)^2 \middle| \psi\right] \tag{15}$$

$$\delta_\tau = \frac{1}{n_\tau}\sum_{j=1}^{G}\sum_{y_{i\tau}^{PIT}=j}\mathbb{E}\left[\lambda_j^{(s+1)}\tilde{y}_{it} - \eta_{it} + \alpha_1^{(s+1)} + \sum_{g=2}^{j-1}\lambda_g^{(s+1)} \middle| y\right], \quad \text{for } \tau \in \{1,\ldots,T\} \tag{16}$$

Note that equation (14) is a quadratic expression in $\lambda_k^{(s+1)}$. Following Ruud (1991), I take the largest root as the update.

In the model without year fixed effects, the update for the fixed effects in equation (16) is replaced by an update for the first threshold α_1:

$$\alpha_1^{(s+1)} = \frac{1}{N}\sum_{j=1}^{G}\sum_{(i,t):y_{it}^{PIT}=j}\mathbb{E}\left[\eta_{it} - \lambda_j^{(s)}\tilde{y}_{it} - \sum_{g=2}^{j-1}\lambda_g^{(s)} \middle| y\right]. \tag{17}$$

3.3 Simulations

To assess the performance of the model, I conduct a simulation study. I generate synthetic data for 3,000 firms, each observed for 20 periods. The data-generating process (DGP) follows the model structure described in Section 3.1, assuming six ordered rating categories. The second column of Table 2 reports the true parameter values used in the simulations.

The model is then estimated using the SEM algorithm described in Section 3.2. I perform 1,000 SEM iterations, using 1,000 MCMC draws in each Expectation step. At each iteration, only the final MCMC draw is retained to approximate the conditional expectation of the complete-data log-likelihood. The final parameter estimates are then obtained by averaging the last 250 iterations.

I replicate the simulation on three different random samples.[8] Results are reported in Table 2. Overall, the model performs well, with estimated parameters generally close to the true values. That said, these comparisons should be interpreted cautiously, as confidence intervals were not computed.

Figure 7 in Appendix A illustrates the convergence behavior of two key variance components: the variance of the idiosyncratic transitory shocks, σ_ε^2, and the variance of the

[8] Ideally, a larger number of replications would provide a more comprehensive assessment of the performance of the model. However, computational constraints limited the number of simulations: estimating a single sample takes approximately one day using 10 cores on CEMFI's server.

Variable	Value	Simulation 1	Simulation 2	Simulation 3
σ_v	0.0375	0.038	0.036	0.037
σ_ε	0.5	0.503	0.489	0.501
λ_1	0.514	0.517	0.508	0.506
λ_2	0.68	0.684	0.668	0.678
λ_3	1.135	1.142	1.122	1.135
λ_4	1.47	1.470	1.452	1.465
δ_1	0.116	0.081	0.099	0.110
δ_2	-0.346	-0.377	-0.351	-0.380
δ_3	-1.194	-1.229	-1.200	-1.197
δ_4	0.316	0.291	0.313	0.290
δ_5	-0.798	-0.852	-0.786	-0.803
δ_6	0.796	0.771	0.790	0.788
δ_7	-0.57	-0.607	-0.580	-0.561
δ_8	-1.014	-1.050	-1.028	-1.026
δ_9	1.2	1.197	1.187	1.203
δ_{10}	-1.119	-1.149	-1.102	-1.138
δ_{11}	-0.14	-0.164	-0.168	-0.165
δ_{12}	-0.829	-0.875	-0.847	-0.837
δ_{13}	-0.344	-0.401	-0.348	-0.363
δ_{14}	-0.459	-0.490	-0.469	-0.473
δ_{15}	1.095	1.071	1.082	1.067
δ_{16}	-1.251	-1.307	-1.256	-1.255
δ_{17}	1.078	1.062	1.031	1.056
δ_{18}	-0.464	-0.530	-0.461	-0.449
δ_{19}	0.309	0.274	0.294	0.277
δ_{20}	-0.616	-0.675	-0.610	-0.629

Table 2: Simulation results. The table reports the true parameter values used to generate the data (second column) and the estimated values obtained from three different random samples.

innovations to the persistent component, σ_ν^2. The plots show the evolution of the estimates across SEM iterations for each of the three samples, where the dashed red lines denote the true values used in the DGP. The results indicate that both variance parameters stabilize after approximately 500 iterations.

4 Empirical Application: TTC Ratings for Spanish Firms

I apply the model to the random subsample of 5,000 firms described in Section 2. The estimation is conducted using 3,000 iterations of the SEM algorithm, with 1,000 MCMC draws per Expectation step. As in the simulation study, I retain only the final MCMC draw from each iteration to approximate the conditional expectation of the complete-data log-likelihood.

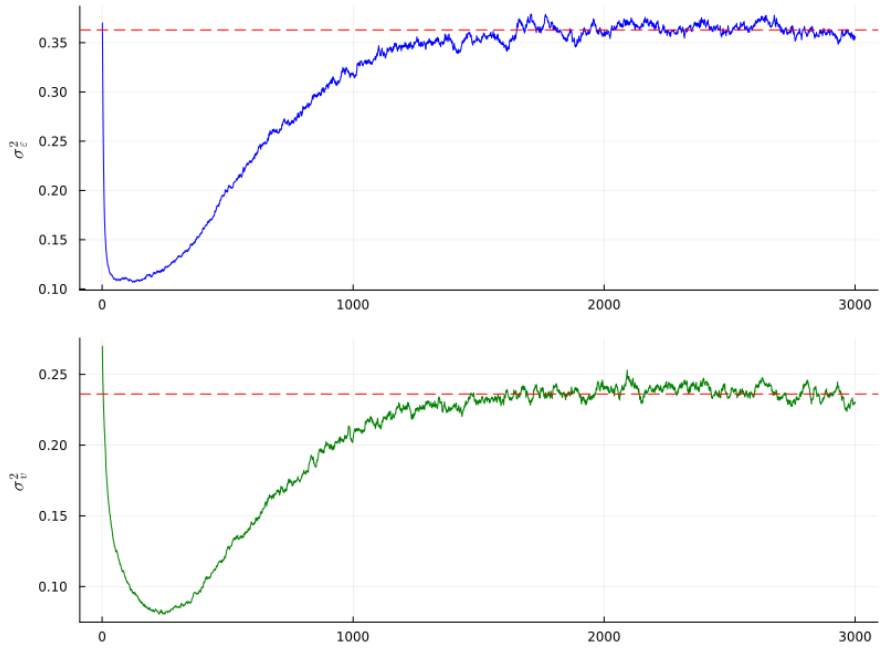


Figure 2: Convergence of parameter estimates in the empirical application. The top panel shows the evolution of the estimated variance of the idiosyncratic transitory shocks (σ^2_ε), while the bottom panel displays the estimated variance of the innovations to the persistent component (σ^2_ν). The dashed red lines represent the final estimates, computed as the average over the last 500 iterations.

The final parameter estimates are then computed as the average over the last 500 iterations. For the initial values, I set all year fixed effects δ_t to zero, the interval lengths λ_j to 0.5, and the variances σ^2_ε and σ^2_ν to 0.37 and 0.27, respectively.[9]

Figure 2 presents the trajectory of the estimated variance components σ^2_ε and σ^2_ν over the course of the 3,000 SEM iterations. The top panel corresponds to the idiosyncratic transitory shocks, and the bottom panel to the innovations in the persistent component. In both cases, the estimates stabilize well before the 2,000th iteration, indicating satisfactory convergence of the algorithm. The dashed red lines denote the final estimates used in the analysis.

Figure 3 plots the estimated year fixed effects δ_t across the sample period. As discussed in Section 3.1, these fixed effects capture the cyclical component embedded in PIT credit ratings. Higher values of δ_t indicate periods in which firms are, on average, assigned worse ratings, consistent with macroeconomic downturns.

The figure reveals clear procyclical dynamics in the ratings. Prior to the Great Financial Crisis, the year fixed effects were relatively low, implying more favorable ratings. Starting in

[9]These initial values were informed by previous estimations on a different subsample of the dataset, consisting of large and medium-sized firms observed in all periods. In that preliminary exercise, the model was first estimated on four random subsamples of 600 firms each, using random initial parameter values. This strategy proved effective, as the average of the four subsample estimates closely approximated the final parameter values, with all four estimates themselves being fairly similar.

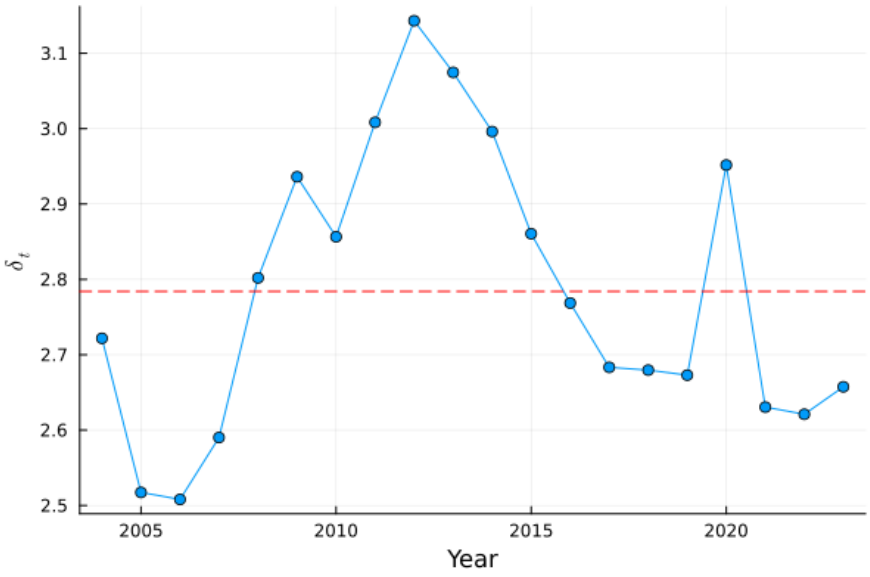


Figure 3: Estimated year fixed effects. The figure displays the estimated year fixed effects δ_t over time. The dashed red line indicates the average value across years. Higher values of δ_t reflect a deterioration in average credit quality.

2008, the year fixed effects increased markedly, reflecting a deterioration in credit quality that continued through the European sovereign debt crisis. The time fixed effects peaked in 2012, consistent with macroeconomic stress during that period.

In the subsequent years, the year fixed effects gradually declined, reaching below-average levels by 2016, suggesting a recovery in firm-level creditworthiness. From 2017 onward, the year fixed effects remained relatively stable, with the exception of a sharp spike in 2020—likely driven by the economic shock caused by the COVID-19 pandemic. This was followed by a rapid normalization in 2021, as captured by a drop in the time fixed effects back toward pre-pandemic levels.

Overall, the estimated year fixed effects provide strong evidence that the original ratings are influenced by macroeconomic conditions, reinforcing their classification as PIT ratings.

Having obtained the TTC ratings, constructed as described in Section 3.1,[10] we can now compare their behavior to that of the original PIT ratings. Table 3 displays the first-order transition matrices for both. Each entry (i, j) represents the estimated probability of transitioning from rating category i to category j in one year.

As expected, the diagonal elements of the TTC transition matrix are noticeably larger than those of the PIT matrix. This indicates that TTC ratings are more persistent, which

[10]The TTC ratings are computed as described in equation 5, based on the conditional expectation $\mathbb{E}[\eta_{it} \mid y_{i1}^{PIT}, \ldots, y_{iT(i)}^{PIT}]$, which is approximated by averaging the last 1,500 MCMC draws after a burn-in period of 3,500 draws.

	4	5+	5	5-	6+	6	6-	7+	7	D	Q
4	**0.74**	0.17	0.01	0.03	0.03	0.01	0	0.01	0	0	0
5+	0.06	**0.63**	0.14	0.02	0.1	0.02	0.01	0.01	0	0	0
5	0.01	0.29	**0.33**	0	0.27	0.04	0.03	0.01	0.01	0	0
5-	0.11	0.26	0.02	**0.2**	0.2	0.1	0.01	0.07	0.02	0	0.01
6+	0.01	0.1	0.17	0.02	**0.46**	0.1	0.08	0.04	0.03	0	0
6	0.01	0.04	0.06	0.02	0.32	**0.21**	0.14	0.11	0.07	0.01	0.01
6-	0	0.02	0.05	0	0.24	0.14	**0.31**	0.13	0.1	0.01	0.01
7+	0.01	0.02	0.02	0.02	0.15	0.13	0.16	**0.32**	0.16	0.01	0.01
7	0	0.01	0.01	0	0.08	0.07	0.11	0.18	**0.49**	0.02	0.02
D	0.01	0.01	0.01	0	0.04	0.02	0.04	0.05	0.06	**0.62**	0.14
Q	0	0	0	0	0	0	0	0	0	0	**1**

	4	5+	5	5-	6+	6	6-	7+	7	D	Q
4	**0.93**	0.07	0	0	0	0	0	0	0	0	0
5+	0.03	**0.87**	0.09	0	0	0	0	0	0	0	0
5	0	0.21	**0.59**	0.05	0.14	0	0	0	0	0	0
5-	0	0.03	0.45	**0.11**	0.41	0	0	0	0	0	0
6+	0	0	0.08	0.03	**0.77**	0.09	0.02	0	0	0	0
6	0	0	0	0	0.26	**0.46**	0.24	0.03	0	0	0
6-	0	0	0	0	0.03	0.19	**0.55**	0.2	0.02	0	0
7+	0	0	0	0	0	0.01	0.18	**0.62**	0.18	0	0
7	0	0	0	0	0	0	0	0.12	**0.83**	0.05	0
D	0	0	0	0	0	0	0	0	0.11	**0.62**	0.27
Q	0	0	0	0	0	0	0	0	0	0	**1**

Table 3: First-order transition matrices for the original PIT (top) and the estimated TTC (bottom) ratings. Rows denote the current category, and columns denote the next category. Entry (i, j) corresponds to the number of transitions from category i to j, divided by the total number of transitions from category i to any category. Table 6 in the Appendix A shows the first-order transition matrix for the original PIT ratings for the full sample.

is consistent with their intended role of smoothing out transitory shocks and macroeconomic fluctuations. In contrast, PIT ratings respond more rapidly to short-term variations, resulting in lower transition persistence[11].

Figure 4 compares the distribution of PIT and TTC ratings in two contrasting macroeconomic periods: 2011, during the European sovereign debt crisis, and 2018, a period of economic expansion. The top panel presents the original PIT ratings, while the bottom panel shows the estimated TTC ratings.

In 2011, PIT ratings exhibit a clear shift toward lower credit categories, reflecting the downturn in aggregate credit quality. The TTC ratings, in contrast, are more evenly dis-

[11] A useful robustness check is to verify that the transitions implied by the model, defined as $\mathbb{E}[y_{it}^* \mid y_{i1}^{PIT}, \ldots, y_{iT(i)}^{PIT}]$, match the observed PIT transitions. This alignment is guaranteed by construction, as the complete-data log-likelihood in equation 12 assigns zero probability to inconsistent combinations of y_{it}^{PIT} and $\tilde{y}_{it}$—i.e., when $\tilde{y}_{it} \notin I_{it}$. Table 7 in the Appendix B confirms this consistency: the transition matrix implied by the fitted PIT model coincides exactly with the upper panel of Table 3.

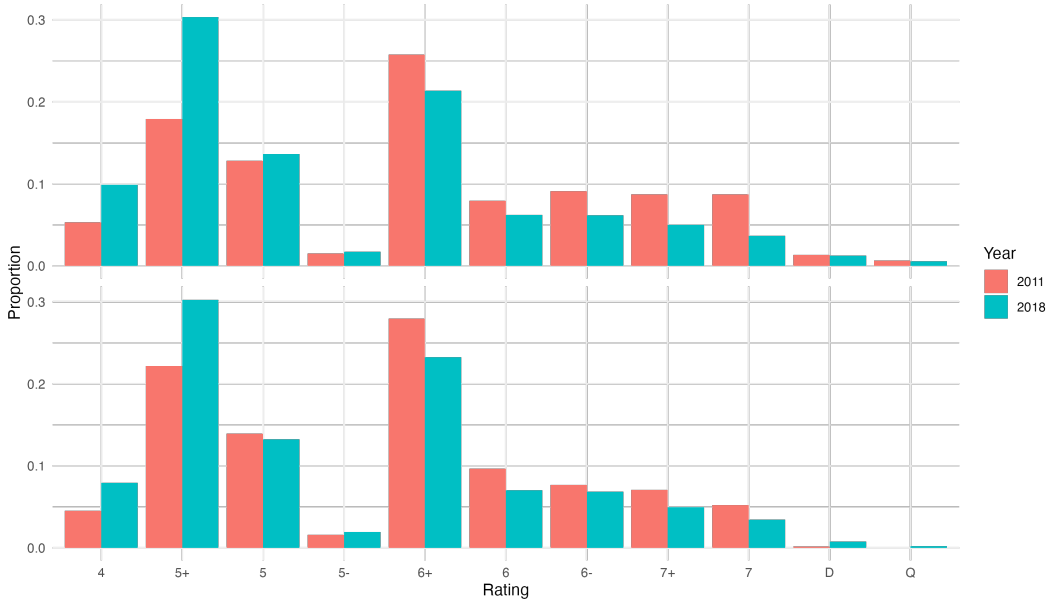


Figure 4: Distribution of the original PIT (top) and the estimated TTC (bottom) ratings in 2011 and 2018. The figure shows the proportion of firms in each rating category. Figure 8 in the Appendix A displays the distribution of the original PIT ratings for the full sample.

tributed and place a greater share of firms in higher-rated categories—most notably in category 5+. This upward shift suggests that TTC ratings successfully smooth over the temporary macroeconomic distress of the recession. For example, the proportion of firms in category 5+ increases by approximately 5 percentage points under the TTC framework.

In 2018, PIT ratings are concentrated in the higher categories, reflecting the improved macroeconomic conditions. However, the TTC ratings moderate this optimism by slightly reducing the share of firms in the top categories (e.g., category 4), again reflecting their resistance to cyclical fluctuations.

Overall, the TTC ratings result in a more stable distribution across the business cycle. They attenuate the cyclical volatility captured by the PIT ratings—raising ratings in recessions and lowering them in expansions—consistent with their goal of isolating firms' long-term creditworthiness from transitory shocks. This behavior aligns with the theoretical expectation that default probabilities rise during downturns and fall during booms, while TTC ratings remain comparatively stable.

Figure 5 displays the time-series evolution of credit ratings for four firms, comparing the original PIT ratings to the TTC ratings estimated using two model variants: one that includes year fixed effects (TTC FE) and one that does not (TTC No FE).[12]

[12]The firms were selected from those observed in more than 10 periods. The top row shows two firms among those with the highest variance in the difference between PIT and TTC ratings, while the bottom row shows

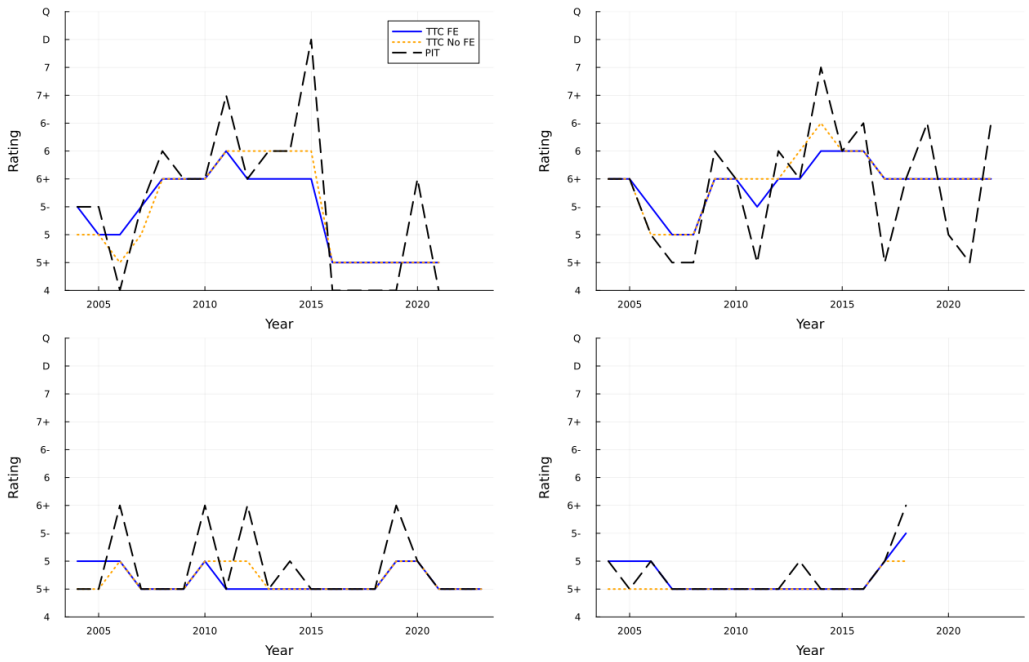


Figure 5: Evolution of the original PIT and the estimated TTC ratings for four firms. The TTC ratings are estimated using two versions of the model: one with year fixed effects (TTC FE) and one without them (TTC No FE).

The figure illustrates the smoothing property of TTC ratings. While PIT ratings fluctuate substantially across years—especially during and after macroeconomic shocks—TTC ratings are more stable, as they abstract from temporary disturbances. This is particularly evident in the top two panels, where PIT ratings exhibit substantial swings, while both TTC variants remain smoother and more persistent.

Comparing the TTC ratings estimated with and without year fixed effects reveals further insights. The TTC FE ratings tend to be worse than the TTC No FE ratings in the pre-crisis period (2005–2007) and better during the crisis years (2008–2015). This pattern reflects the role of year fixed effects in capturing the cyclical component embedded in PIT ratings. By accounting for aggregate shocks, the TTC FE ratings provide a cleaner measure of firms' underlying creditworthiness across the business cycle—penalizing firms less in downturns and moderating optimism in booms.

Future Extensions

A natural next step would be to replace the smoothing approach used to estimate the persistent component with a filtering approach, enabling real-time TTC rating updates. One option would be to re-estimate the model sequentially using an expanding window strat-

two firms among those with the highest variance in the difference between the TTC FE and TTC No FE ratings (they were chosen to enhance interpretability, avoiding excessively volatile or flat time series).

egy—estimating the model first with 11 periods, then with 12, and so on. However, this is computationally demanding. A more feasible alternative is to estimate the model once using the full sample, and then re-estimate the latent persistent component using a filtering procedure conditioned on the fixed parameters. Ideally, future work would explore the use of Sequential Monte Carlo (SMC) methods, which are better suited for filtering in dynamic latent variable models and offer a principled alternative to the MCMC-based smoother used in this thesis.

Another promising extension involves scaling up the analysis to include all firms in the dataset, rather than a random subsample of 5,000 firms. A practical way to reduce computational demands would be to estimate the model parameters using a smaller random subset of firms and subsequently compute TTC ratings for the entire sample based on those estimated parameters.

A particularly interesting avenue would be to use a different random subsample of firms in the Expectation step at each iteration. This would significantly reduce computational costs, as the Expectation step is the most computationally intensive part of the algorithm, while still allowing the use of all observations for estimation. Since the Maximization step only requires the average of the firm-level conditional expectations, this average might be reliably approximated from a smaller random subset. This strategy would resemble the logic of Stochastic Gradient Descent and may allow the model to scale efficiently to very large datasets.

5 Conclusion

This thesis introduces a novel method to estimate TTC ratings from PIT ratings by decomposing the latter into a persistent component and a transitory component. Importantly, the proposed approach leverages the panel structure of the data to account for unobserved heterogeneity in firms' credit quality. This allows the model to flexibly capture both transitory idiosyncratic dynamics and cyclical macroeconomic effects through the inclusion of year fixed effects.

I apply the proposed method to a novel dataset of firm-level credit ratings used by the Bank of Spain to assess the credit risk of Spanish non-financial firms for monetary policy operations. The resulting TTC ratings exhibit the desired smoothing properties: they attenuate not only the cyclical volatility present in PIT ratings but also transitory fluctuations in creditworthiness—an aspect often overlooked in previous attempts to derive TTC ratings

from PIT measures. At the same time, these TTC ratings remain responsive to long-term structural shifts in firms' credit profiles.

The estimated year fixed effects uncover a strong procyclical pattern in PIT ratings. These effects are significantly elevated during major downturns such as the Great Financial Crisis, the European sovereign debt crisis, and the COVID-19 pandemic in 2020, indicating systematically lower PIT ratings during these periods. A comparison between TTC ratings estimated with and without year fixed effects further confirms that incorporating these effects allows the model to correct for overly optimistic ratings during expansions and overly pessimistic ratings during recessions.

A promising direction for future research is to replace the current smoothing framework with a filtering approach for estimating the persistent component in real time. In particular, the application of Sequential Monte Carlo methods—rather than the Markov Chain Monte Carlo methods employed in this thesis—offers an appealing avenue to enhance the model's applicability in dynamic settings.

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Appendix

A Additional Tables and Figures

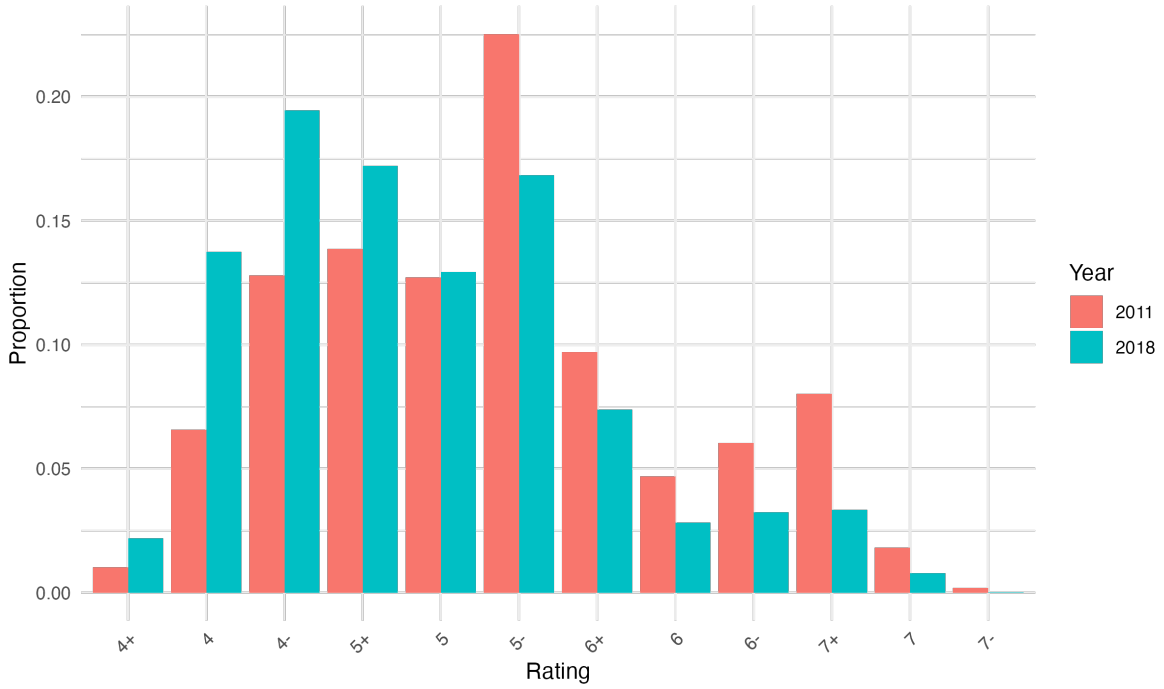


Figure 6: Distribution of credit ratings in 2011 and 2018 for large firms. The figure shows the proportion of large firms in each rating category.

4	5+	5	5-	6+	6	6-	7+	7	D	Q
3777	12677	7590	1074	14118	4797	4907	4227	3914	585	324

Table 4: Number of firm-year observations in each rating category for the random subsample. The table shows the number of observations in each rating category, including the absorbing states for default (D) and bankruptcy (Q).

4	5+	5	5-	6+	6	6-	7+	7	D	Q
735672	3075350	1988740	235397	4367390	1627285	1737552	1481490	1722935	207536	120636

Table 5: Number of firm-year observations in each rating category for the whole sample. The table shows the number of observations in each rating category, including the absorbing states for default (D) and bankruptcy (Q).

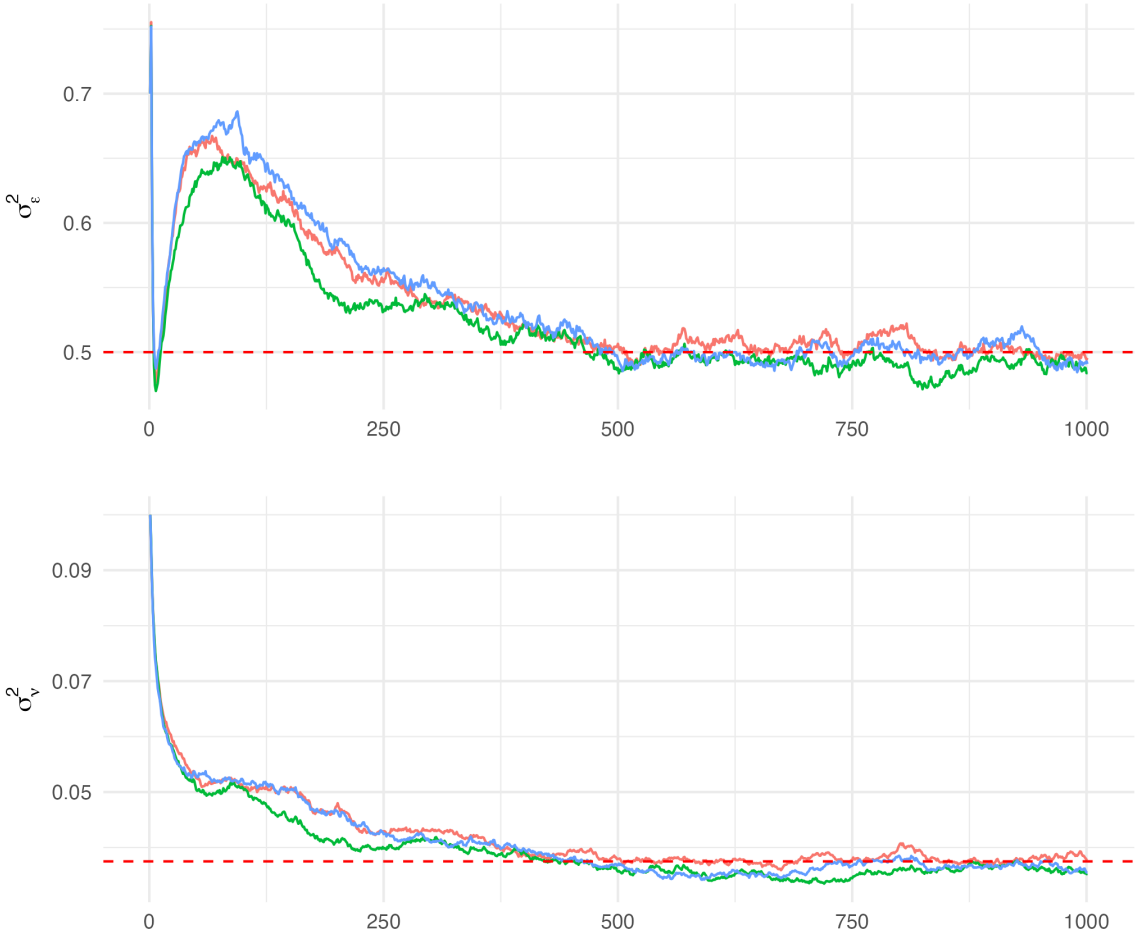


Figure 7: Convergence of parameter estimates in the simulations. The plots show the evolution of the estimated variances of the idiosyncratic transitory shocks (σ^2_ε, top panel) and the innovations to the persistent component (σ^2_ν, bottom panel) over 1,000 SEM iterations, across three random samples. The dashed red lines indicate the true values.

	4	5+	5	5-	6+	6	6-	7+	7	D	Q
4	**0.74**	0.16	0.01	0.03	0.03	0.02	0	0.01	0	0	0
5+	0.06	**0.62**	0.15	0.02	0.11	0.02	0.01	0.01	0	0	0
5	0	0.32	**0.31**	0	0.26	0.04	0.03	0.01	0.01	0	0
5-	0.14	0.28	0.02	**0.2**	0.18	0.08	0.01	0.06	0.02	0	0.01
6+	0.01	0.13	0.17	0.01	**0.44**	0.09	0.07	0.04	0.03	0	0
6	0.01	0.05	0.08	0.02	0.36	**0.18**	0.13	0.09	0.06	0.01	0.01
6-	0	0.03	0.06	0	0.26	0.15	**0.28**	0.12	0.09	0.01	0.01
7+	0.01	0.02	0.03	0.01	0.17	0.12	0.18	**0.3**	0.15	0.01	0.01
7	0	0.01	0.02	0	0.1	0.07	0.12	0.19	**0.46**	0.02	0.02
D	0	0.02	0.01	0	0.05	0.02	0.03	0.03	0.05	**0.61**	0.16
Q	0	0	0	0	0	0	0	0	0	0	**1**

Table 6: First-order transition matrix for the original PIT ratings using the whole sample. Rows denote the current category, and columns denote the next category. Entry (i, j) corresponds to the number of transitions from category i to j, divided by the total number of transitions from category i to any category.

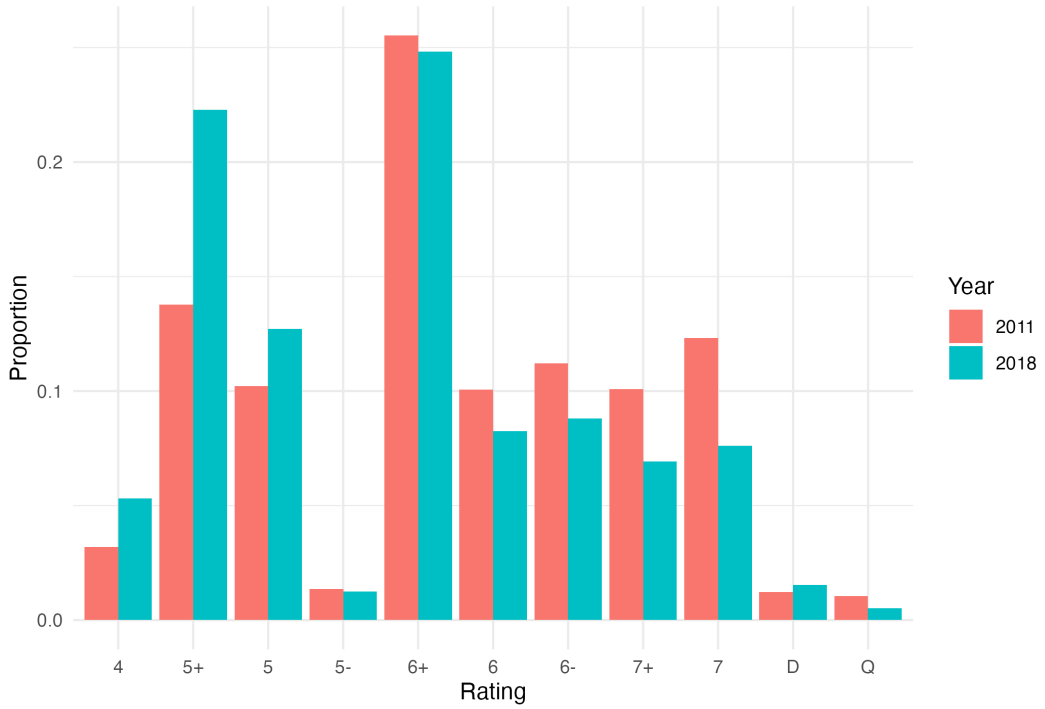


Figure 8: Distribution of the original PIT ratings in 2011 and 2018 using the whole sample. The figure shows the proportion of firms in each rating category.

B Implied Transitions

Given the estimated parameters, we can approximate the conditional expectation of the latent variables $\tilde{y}_{it}$ and η_{it} given the observed ratings using the Metropolis-Hastings algorithm employed in the Expectation step. To analyze the transitions implied by the model, one has to first undo the transformation applied to the latent variable y^*_{it}:

$$\mathbb{E}[y^*_{it} \mid y^{PIT}_{i1}, \ldots, y^{PIT}_{iT(i)}] = \alpha_{j-1} + \lambda_j \mathbb{E}[\tilde{y}_{it} \mid y^{PIT}_{i1}, \ldots, y^{PIT}_{iT(i)}], \quad \text{if } y^{PIT}_{it} = j \text{ and } j \in \{2, \ldots, G-1\},$$
$$\mathbb{E}[y^*_{it} \mid y^{PIT}_{i1}, \ldots, y^{PIT}_{iT(i)}] = \alpha_1 + \mathbb{E}[\tilde{y}_{it} \mid y^{PIT}_{i1}, \ldots, y^{PIT}_{iT(i)}], \quad \text{if } y^{PIT}_{it} = 1,$$
$$\mathbb{E}[y^*_{it} \mid y^{PIT}_{i1}, \ldots, y^{PIT}_{iT(i)}] = \alpha_{G-1} + \mathbb{E}[\tilde{y}_{it} \mid y^{PIT}_{i1}, \ldots, y^{PIT}_{iT(i)}], \quad \text{if } y^{PIT}_{it} = G.$$

Applying this transformation to the estimated conditional expectations of $\tilde{y}_{it}$, we can compute the transitions implied by the model. The resulting transition matrix is shown in Table 7, which coincides with the transitions in the upper panel of Table 3. Note that this holds by construction, as the complete-data log-likelihood in equation 12 assigns zero probability to inconsistent combinations of y^{PIT}_{it} and $\tilde{y}_{it}$—i.e., when $\tilde{y}_{it} \notin I_{it}$.

	4	5+	5	5-	6+	6	6-	7+	7	D	Q
4	**0.74**	0.17	0.01	0.03	0.03	0.01	0	0.01	0	0	0
5+	0.06	**0.63**	0.14	0.02	0.1	0.02	0.01	0.01	0	0	0
5	0.01	0.29	**0.33**	0	0.27	0.04	0.03	0.01	0.01	0	0
5-	0.11	0.26	0.02	**0.2**	0.2	0.1	0.01	0.07	0.02	0	0.01
6+	0.01	0.1	0.17	0.02	**0.46**	0.1	0.08	0.04	0.03	0	0
6	0.01	0.04	0.06	0.02	0.32	**0.21**	0.14	0.11	0.07	0.01	0.01
6-	0	0.02	0.05	0	0.24	0.14	**0.31**	0.13	0.1	0.01	0.01
7+	0.01	0.02	0.02	0.02	0.15	0.13	0.16	**0.32**	0.16	0.01	0.01
7	0	0.01	0.01	0	0.08	0.07	0.11	0.18	**0.49**	0.02	0.02
D	0.01	0.01	0.01	0	0.04	0.02	0.04	0.05	0.06	**0.62**	0.14
Q	0	0	0	0	0	0	0	0	0	0	**1**

Table 7: First-order transition matrix implied by the estimated PIT model. This table shows the migration probabilities implied by the fitted PIT model, which coincides with the upper panel of Table 3. Rows denote the current category, and columns denote the next category. Entry (i, j) corresponds to the number of transitions from category i to j, divided by the total number of transitions from category i to any category.